

# PROS9-A1890-OSB-M

Serie PROS-M · R134A · senza serbatoio / without tank



PROS9-A1890-OSB-M

RESA FRIGORIFERA · COOLING CAPACITY

**1.242** kW

EER · 12/7 °C · AIR 35 °C

**2,99**

RESA NOMINALE · 15 °C · AIR 25 °C

**1.887** kW

PORTATA ACQUA · WATER FLOW

**213,5** m<sup>3</sup>/h

|                            |                   |                                        |              |
|----------------------------|-------------------|----------------------------------------|--------------|
| Compressori / Compressors  | 2 × vite          | L × P × H (mm)                         | su richiesta |
| Circuiti / Circuits        | 2                 | Serbatoio / Tank                       | —            |
| Ventilatori / Fans         | 26                | Taglia / Size                          | 1890         |
| Refrigerante / Refrigerant | R134A · GWP 1.430 | Campo aria esterna / Ambient air range | -5 / +50 °C  |

## Resa in sintesi / Capacity at a glance <sup>kW</sup>

| ARIA ESTERNA / AMBIENT AIR                  | 25 °C | 30 °C | 35 °C | 40 °C | 45 °C | 50 °C |
|---------------------------------------------|-------|-------|-------|-------|-------|-------|
| Acqua in uscita 7 °C / Leaving water 7 °C   | 1.384 | 1.314 | 1.242 | 1.168 | 1.092 | 1.015 |
| Acqua in uscita 15 °C / Leaving water 15 °C | 1.887 | 1.796 | 1.702 | 1.605 | 1.507 | 1.406 |

Refrigeratori condensati ad aria con compressori a vite e condensatore a microcanali, selezionabili con acqua in uscita fino a 17 °C: indicati per i processi che lavorano con acqua non troppo fredda.

Air-cooled chillers with screw compressors and micro-channel condenser, selectable with leaving water up to 17 °C: suited to processes that run on water that is not too cold.

# Dati tecnici

## Technical data

PROS9 - A1890 - OSB - M

**Prestazioni nominali — acqua in uscita 15 °C, aria 25 °C / Nominal performance — 15 °C leaving water, 25 °C air**

|                                     |                          |
|-------------------------------------|--------------------------|
| Resa frigorifera / Cooling capacity | 1.887,20 kW              |
| Resa frigorifera / Cooling capacity | 1.622.992 kcal/h         |
| Potenza assorbita / Power input     | 353,00 kW                |
| EER / EER                           | 5,35                     |
| Portata acqua / Water flow          | 324,60 m <sup>3</sup> /h |

**Condizioni standard — acqua 12/7 °C, aria 35 °C / Standard conditions — 12/7 °C water, 35 °C air**

|                                     |                          |
|-------------------------------------|--------------------------|
| Resa frigorifera / Cooling capacity | 1.241,50 kW              |
| Potenza assorbita / Power input     | 414,73 kW                |
| EER / EER                           | 2,99                     |
| ESEER / ESEER                       | 3,94                     |
| Portata acqua / Water flow          | 213,54 m <sup>3</sup> /h |

### Componenti / Components

|                                             |                        |
|---------------------------------------------|------------------------|
| Compressori / Compressors                   | 2 × Vite               |
| Circuiti frigoriferi / Refrigerant circuits | 2                      |
| Ventilatori / Fans                          | 26                     |
| Condensatore / Condenser                    | a microcanali          |
| Evaporatore / Evaporator                    | a fascio tubiero       |
| Refrigerante / Refrigerant                  | R134A • GWP 1.430 • A1 |

### Installazione / Installation

|                                                                                           |                           |
|-------------------------------------------------------------------------------------------|---------------------------|
| Dimensioni L × P × H / Dimensions L × D × H                                               | su richiesta / on request |
| Campo aria esterna / Ambient air range                                                    | -5 / +50 °C               |
| Dati elettrici, carica gas, peso, attacchi / Electrical data, charge, weight, connections | su richiesta / on request |

### Dotazioni di serie / Standard equipment

- Valvole di intercettazione  
Shut-off valves
- Griglia di protezione condensatore  
Condenser protection grille
- Valvola di espansione elettronica  
Electronic expansion valve
- Condensatore a microcanali  
Micro-channel condenser
- Evaporatore a fascio tubiero  
Shell & tube evaporator
- Display 4"/7"  
4"/7" display
- Pressostati e manometri  
Pressure switches and gauges
- Pressostato differenziale acqua  
Water differential pressure switch
- Presa di servizio 220 V e luce quadro elettrico  
220 V service socket and electrical panel light

### Con glicole / With glycol

| MISCELA  | COEFF.  | 12/7 • 35 °C |
|----------|---------|--------------|
| MEG 20 % | × 0,980 | 1.216,7 kW   |
| MEG 30 % | × 0,950 | 1.179,4 kW   |
| MEG 40 % | × 0,920 | 1.142,2 kW   |
| MPG 20 % | × 0,975 | 1.210,5 kW   |
| MPG 30 % | × 0,960 | 1.191,8 kW   |
| MPG 40 % | × 0,950 | 1.179,4 kW   |

# Prestazioni Performance

PROS9-A1890-OSB-M · acqua pura,  $\Delta T$  5 K · pure water, 5 K  $\Delta T$  ·  $\square$  12/7 °C · 35 °C

## Resa frigorifera / Cooling capacity (kW)

| LWT ↓ · air → | 15    | 20    | 25    | 30    | 35    | 40    | 45    | 50    |
|---------------|-------|-------|-------|-------|-------|-------|-------|-------|
| 1             | 1.188 | 1.135 | 1.079 | 1.023 | 964   | 904   | 843   | 781   |
| 2             | 1.240 | 1.184 | 1.127 | 1.068 | 1.008 | 945   | 882   | 818   |
| 3             | 1.291 | 1.234 | 1.174 | 1.113 | 1.051 | 986   | 921   | 854   |
| 4             | 1.346 | 1.287 | 1.225 | 1.162 | 1.097 | 1.030 | 962   | 893   |
| 5             | 1.401 | 1.339 | 1.276 | 1.210 | 1.143 | 1.074 | 1.003 | 932   |
| 6             | 1.459 | 1.396 | 1.330 | 1.262 | 1.192 | 1.121 | 1.048 | 973   |
| 7             | 1.517 | 1.452 | 1.384 | 1.314 | 1.242 | 1.168 | 1.092 | 1.015 |
| 8             | 1.579 | 1.512 | 1.441 | 1.369 | 1.294 | 1.217 | 1.139 | 1.059 |
| 9             | 1.642 | 1.571 | 1.498 | 1.424 | 1.346 | 1.267 | 1.186 | 1.104 |
| 10            | 1.708 | 1.635 | 1.560 | 1.482 | 1.402 | 1.320 | 1.236 | 1.151 |
| 11            | 1.774 | 1.698 | 1.621 | 1.540 | 1.458 | 1.373 | 1.286 | 1.198 |
| 12            | 1.844 | 1.766 | 1.685 | 1.602 | 1.517 | 1.429 | 1.340 | 1.249 |
| 13            | 1.914 | 1.833 | 1.750 | 1.664 | 1.576 | 1.486 | 1.393 | 1.299 |
| 14            | 1.988 | 1.905 | 1.819 | 1.730 | 1.639 | 1.545 | 1.450 | 1.352 |
| 15            | 2.062 | 1.976 | 1.887 | 1.796 | 1.702 | 1.605 | 1.507 | 1.406 |
| 16            | 2.140 | 2.052 | 1.960 | 1.866 | 1.768 | 1.669 | 1.567 | 1.463 |
| 17            | 2.218 | 2.127 | 2.033 | 1.935 | 1.835 | 1.732 | 1.627 | 1.520 |

## Potenza assorbita / Power input (kW)

| LWT ↓ · air → | 15  | 20  | 25  | 30  | 35  | 40  | 45  | 50  |
|---------------|-----|-----|-----|-----|-----|-----|-----|-----|
| 1             | 282 | 309 | 339 | 371 | 407 | 447 | 491 | 540 |
| 2             | 283 | 310 | 340 | 372 | 409 | 449 | 493 | 542 |
| 3             | 284 | 311 | 341 | 374 | 410 | 450 | 495 | 544 |
| 4             | 285 | 312 | 342 | 375 | 411 | 452 | 496 | 545 |
| 5             | 286 | 313 | 343 | 376 | 412 | 453 | 498 | 547 |
| 6             | 287 | 314 | 344 | 377 | 414 | 454 | 499 | 548 |
| 7             | 288 | 315 | 345 | 378 | 415 | 455 | 500 | 550 |
| 8             | 289 | 316 | 346 | 379 | 416 | 457 | 502 | 552 |
| 9             | 290 | 317 | 347 | 380 | 417 | 458 | 503 | 553 |
| 10            | 291 | 318 | 348 | 381 | 418 | 459 | 504 | 554 |
| 11            | 292 | 319 | 349 | 382 | 419 | 460 | 506 | 556 |
| 12            | 294 | 320 | 350 | 383 | 420 | 461 | 507 | 557 |
| 13            | 295 | 321 | 351 | 384 | 421 | 462 | 508 | 558 |
| 14            | 296 | 322 | 352 | 385 | 422 | 463 | 509 | 560 |
| 15            | 297 | 323 | 353 | 386 | 423 | 464 | 510 | 561 |
| 16            | 299 | 325 | 354 | 387 | 424 | 465 | 511 | 562 |
| 17            | 300 | 326 | 355 | 388 | 425 | 466 | 512 | 563 |

## EER / EER (kW/kW)

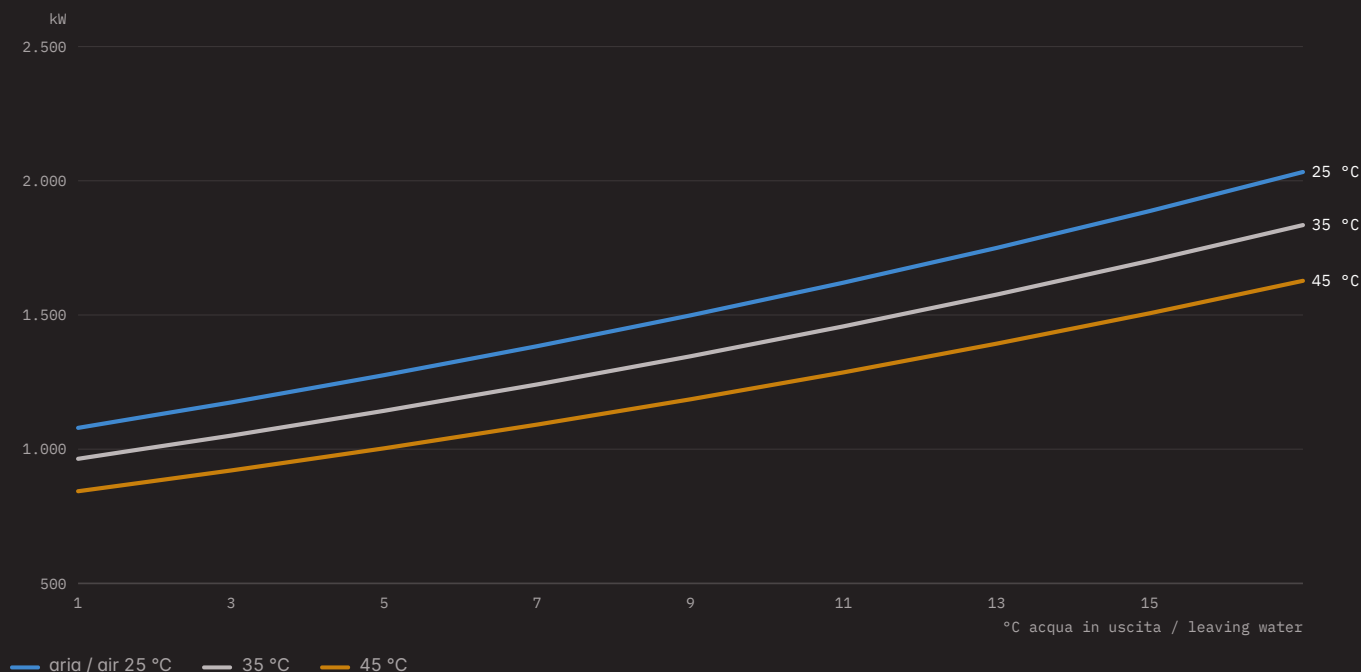
| LWT ↓ · air → | 15   | 20   | 25   | 30   | 35   | 40   | 45   | 50   |
|---------------|------|------|------|------|------|------|------|------|
| 1             | 4,21 | 3,67 | 3,19 | 2,75 | 2,37 | 2,02 | 1,72 | 1,45 |
| 2             | 4,38 | 3,82 | 3,32 | 2,87 | 2,46 | 2,11 | 1,79 | 1,51 |
| 3             | 4,54 | 3,97 | 3,45 | 2,98 | 2,56 | 2,19 | 1,86 | 1,57 |
| 4             | 4,72 | 4,12 | 3,59 | 3,10 | 2,67 | 2,28 | 1,94 | 1,64 |
| 5             | 4,89 | 4,28 | 3,72 | 3,22 | 2,77 | 2,37 | 2,02 | 1,70 |
| 6             | 5,08 | 4,44 | 3,87 | 3,35 | 2,88 | 2,47 | 2,10 | 1,77 |
| 7             | 5,27 | 4,61 | 4,01 | 3,48 | 2,99 | 2,56 | 2,18 | 1,85 |
| 8             | 5,46 | 4,78 | 4,17 | 3,61 | 3,11 | 2,67 | 2,27 | 1,92 |
| 9             | 5,65 | 4,96 | 4,32 | 3,75 | 3,23 | 2,77 | 2,36 | 2,00 |
| 10            | 5,86 | 5,14 | 4,49 | 3,89 | 3,36 | 2,88 | 2,45 | 2,08 |
| 11            | 6,06 | 5,32 | 4,65 | 4,03 | 3,48 | 2,98 | 2,54 | 2,16 |
| 12            | 6,28 | 5,52 | 4,82 | 4,18 | 3,61 | 3,10 | 2,64 | 2,24 |
| 13            | 6,49 | 5,71 | 4,99 | 4,33 | 3,74 | 3,21 | 2,74 | 2,33 |
| 14            | 6,72 | 5,91 | 5,17 | 4,49 | 3,88 | 3,34 | 2,85 | 2,42 |
| 15            | 6,93 | 6,11 | 5,35 | 4,65 | 4,02 | 3,46 | 2,95 | 2,51 |
| 16            | 7,17 | 6,32 | 5,53 | 4,82 | 4,17 | 3,59 | 3,07 | 2,60 |
| 17            | 7,40 | 6,53 | 5,72 | 4,99 | 4,32 | 3,71 | 3,18 | 2,70 |

# Curve

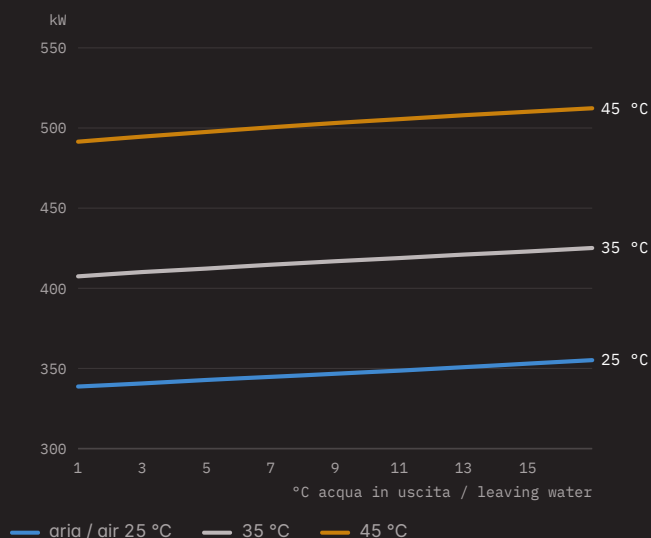
## Curves

PROS9-A1890-OSB-M · acqua pura,  $\Delta T$  5 K · pure water, 5 K  $\Delta T$

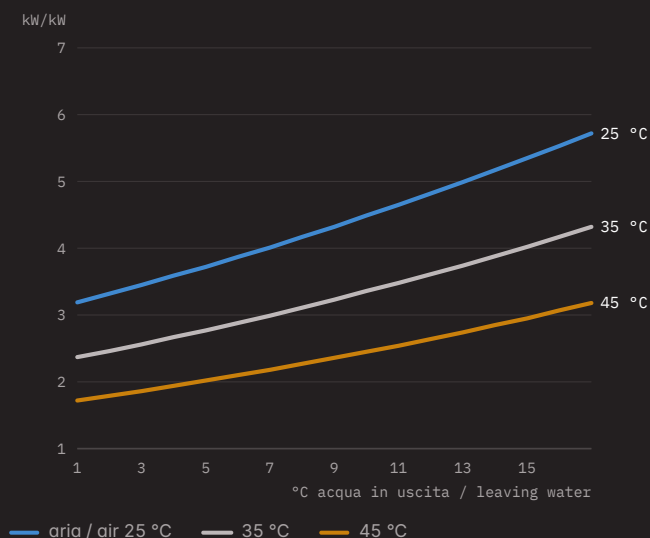
### Resa frigorifera / Cooling capacity



### Potenza assorbita / Power input



### EER / EER



Dimensioni, pesi e disegno d'ingombro sono forniti su richiesta per la configurazione scelta. / Dimensions, weights and dimensional drawing are available on request for the chosen configuration.

# Optional — 1/2 Options

PROS9-A1890-OSB-M · 56 optional disponibili per questa configurazione · 56 options available for this configuration

| Pompe / Pumps                                                                                       |
|-----------------------------------------------------------------------------------------------------|
| Pompa 214 m³/h – prevalenza 25 m<br>Pump 214 m³/h – head 25 m                                       |
| Pompa 214 m³/h – prevalenza 40 m<br>Pump 214 m³/h – head 40 m                                       |
| Pompa doppia montata 214 m³/h – prevalenza 25 m<br>Twin pump, factory -fitted, 214 m³/h – head 25 m |
| Pompa doppia montata 214 m³/h – prevalenza 40 m<br>Twin pump, factory -fitted, 214 m³/h – head 40 m |

| Serbatoi e vasi di espansione / Tanks and expansion vessels                                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Serbatoio idraulico esterno 3000 l in AISI 304 con pompa doppia in AISI 304 montata<br>External water tank 3000 l, AISI 304 with factory-fitted twin AISI 304 pump                         |
| Serbatoio idraulico esterno 5000 l in AISI 304 con pompa doppia in AISI 304 montata<br>External water tank 5000 l, AISI 304 with factory-fitted twin AISI 304 pump                         |
| Serbatoio idraulico esterno 3000 l in AISI 304 con pompa doppia montata<br>External water tank 3000 l, AISI 304 with factory-fitted twin pump                                              |
| Serbatoio idraulico esterno 5000 l in AISI 304 con pompa doppia montata<br>External water tank 5000 l, AISI 304 with factory-fitted twin pump                                              |
| Serbatoio idraulico esterno 3000 l in AISI 304 con pompa singola in AISI 304 montata<br>External water tank 3000 l, AISI 304 with factory-fitted single AISI 304 pump                      |
| Serbatoio idraulico esterno 5000 l in AISI 304 con pompa singola in AISI 304 montata<br>External water tank 5000 l, AISI 304 with factory-fitted single AISI 304 pump                      |
| Serbatoio idraulico esterno 3000 l in AISI 304 con pompa singola montata<br>External water tank 3000 l, AISI 304 with factory-fitted single pump                                           |
| Serbatoio idraulico esterno 5000 l in AISI 304 con pompa singola montata<br>External water tank 5000 l, AISI 304 with factory-fitted single pump                                           |
| Serbatoio idraulico esterno 3000 l pressurizzato in acciaio zincato con pompa singola montata<br>External water tank 3000 l, pressurised, galvanised steel with factory-fitted single pump |
| Serbatoio idraulico esterno 5000 l pressurizzato in acciaio zincato con pompa singola montata<br>External water tank 5000 l, pressurised, galvanised steel with factory-fitted single pump |
| Vaso di espansione 100 l<br>Expansion vessel 100 l                                                                                                                                         |
| Vaso di espansione 200 l<br>Expansion vessel 200 l                                                                                                                                         |
| Vaso di espansione 300 l<br>Expansion vessel 300 l                                                                                                                                         |
| Vaso di espansione 500 l<br>Expansion vessel 500 l                                                                                                                                         |
| Vaso di espansione 750 l<br>Expansion vessel 750 l                                                                                                                                         |

| Valvole e filtri / Valves and filters                    |
|----------------------------------------------------------|
| Elettrovalvola 2"<br>Solenoid valve 2"                   |
| Elettrovalvola 3"<br>Solenoid valve 3"                   |
| Elettrovalvola 4"<br>Solenoid valve 4"                   |
| Elettrovalvola 5"<br>Solenoid valve 5"                   |
| Elettrovalvola 6"<br>Solenoid valve 6"                   |
| Filtro a dischi in acciaio 2"<br>Steel disc filter 2"    |
| Filtro a dischi in acciaio 3"<br>Steel disc filter 3"    |
| Filtro a dischi in acciaio 4"<br>Steel disc filter 4"    |
| Filtro meccanico a Y 2"<br>Y-strainer 2"                 |
| Filtro meccanico a Y 3"<br>Y-strainer 3"                 |
| Filtro meccanico a Y 4"<br>Y-strainer 4"                 |
| Filtro meccanico a Y 5"<br>Y-strainer 5"                 |
| Filtro meccanico a Y 6"<br>Y-strainer 6"                 |
| Valvola a 3 vie 2" miscelatrice<br>3-way mixing valve 2" |
| Valvola a 3 vie 3" miscelatrice<br>3-way mixing valve 3" |
| Valvola a 3 vie 4" miscelatrice<br>3-way mixing valve 4" |
| Valvola a 3 vie 5" miscelatrice<br>3-way mixing valve 5" |
| Valvola a 3 vie 6" miscelatrice<br>3-way mixing valve 6" |
| Valvola a 3 vie 2" on/off<br>3-way valve 2" on/off       |
| Valvola a 3 vie 3" on/off<br>3-way valve 3" on/off       |
| Valvola a 3 vie 4" on/off<br>3-way valve 4" on/off       |
| Valvola a 3 vie 5" on/off<br>3-way valve 5" on/off       |

# Optional — 2/2

## Options

PROS9-A1890-OSB-M · 56 optional disponibili per questa configurazione · 56 options available for this configuration

### Valvole e filtri / Valves and filters

Valvola a 3 vie 6" on/off  
3-way valve 6" on/off

### Circuito frigorifero / Refrigerant circuit

Cofanatura insonorizzante compressori  
Compressor acoustic enclosure

Compressore inverter  
Inverter compressor

Desurriscaldatore a piastre  
Plate desuperheater

Rilevatore fughe gas  
Refrigerant leak detector

### Free cooling / Free cooling

Free cooling 1200 kW – 24 ventilatori (rif. 18)  
Free cooling 1200 kW – 24 fans (ref. 18)

### Ventilazione e clima / Fans and climate

Inverter ventilatori  
Fan inverter

Ventilatori EC  
EC fans

### Controllo e comunicazione / Control and communication

Comando remoto con display (cavo escluso)  
Remote control with display (cable not included)

Comunicazione RS485 Modbus RTU / TCP-IP (cavo escluso)  
RS485 Modbus RTU / TCP-IP communication (cable not included)

Comunicazione RS485 verso BACnet (cavo escluso)  
RS485 to BACnet communication (cable not included)

### Quadro elettrico / Electrical panel

Analizzatore di rete  
Power analyser

Kit invernale (inverter ventilatori + resistenza quadro + resistenza evaporatore + termostato) fino a -25 °C  
Winter kit (fan inverter + panel heater + evaporator heater + thermostat) down to -25 °C

### Installazione / Installation

Antivibranti a molla  
Spring anti-vibration mounts

Antivibranti in gomma  
Rubber anti-vibration mounts

## Note Notes

- Prestazioni con acqua pura, salto termico 5 K. Le condizioni di riferimento sono indicate in ogni tabella.
  - Con miscele di glicole la resa frigorifera si corregge con i coefficienti della tabella dedicata; potenza assorbita ed EER restano invariati.
  - Dimensioni ricavate dai disegni quotati (tolleranza  $\pm 10$  mm): da confermare in fase d'ordine.
  - Dati elettrici, carica di refrigerante, pesi e attacchi idraulici sono forniti su richiesta per la configurazione scelta.
  - Procryon si riserva di modificare dati e caratteristiche senza preavviso, per il miglioramento continuo del prodotto.
- Performance with pure water, 5 K temperature difference. Reference conditions are stated in each table.
  - With glycol mixtures, cooling capacity is corrected with the coefficients in the dedicated table; power input and EER are unchanged.
  - Dimensions taken from dimensional drawings (tolerance  $\pm 10$  mm): to be confirmed at order stage.
  - Electrical data, refrigerant charge, weights and water connections are available on request for the chosen configuration.
  - Procryon reserves the right to change data and features without notice, as part of continuous product improvement.

## Parliamoci Let's talk

Progettazione, intervento urgente, valutazione di un impianto esistente, supporto tecnico e intervento rapido — 24/7.

System design, urgent response, assessment of an existing plant, technical support and rapid response — 24/7.

**Procryon — Smart Cooling Engineering S.r.l.**  
Via Ugo Vaglia 10 — Gavardo (BS) — Italia  
[www.procryon.com](http://www.procryon.com)



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Continuity by design